chhat छत

Solar, without the sales pitch.

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The independent AI solar advisor for every Indian roof.

Team 404 not found HackIndia · AI-First Startup Hackathon 2026

Track: AI in any startup (climate) + FinTech

THE PROBLEM

India installs 16,000 rooftops a day. Nobody is on the homeowner's side.

51 lakh

PM Surya Ghar homes by Sep 2026, half the 1 crore target

5%

of households willing to install, though the economics work (CEEW)

Jul 2026

MNRE acknowledges wrong DISCOM bills after rooftop installs

Every calculator a homeowner finds belongs to someone selling panels.

One quote. Three problems. ₹1.6 lakh at risk.

“3 kW” on-grid system	₹2,41,256
5 × 540 Wp, imported	2.70 kWp
GST @ 13.8%	₹29,256

- **₹57,710** above the fair price band
- **₹72,600** subsidy lost to non-DCR panels
- **0.3 kW** paid for and never installed
- Then after install: **₹2,714** overbilled in one cycle when the DISCOM ignores export

Four checks between a family and a bad deal.

Roof Report

Bill + 20 years of NASA sunlight → size, generation, subsidy, loan, 25-year payback.

Quote Check

14 checks: fair price, DCR and ALMM, capacity, inverter, warranties, GST, payment terms.

Bill Audit

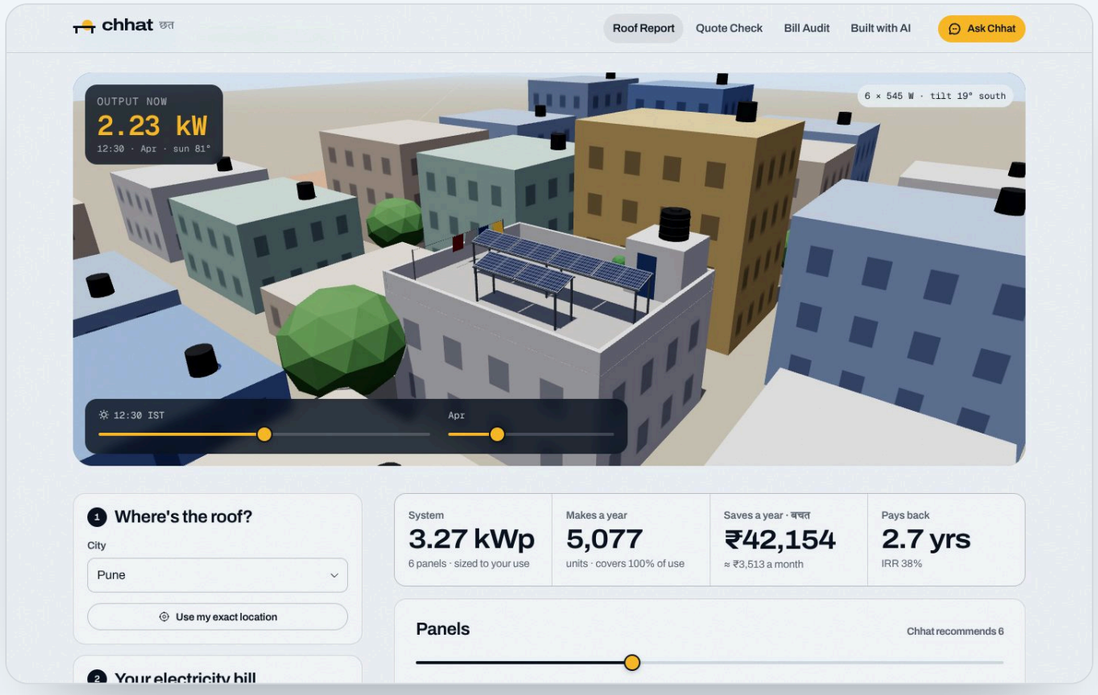
Import – export – credit vs. billed units. Drafts the DISCOM letter.

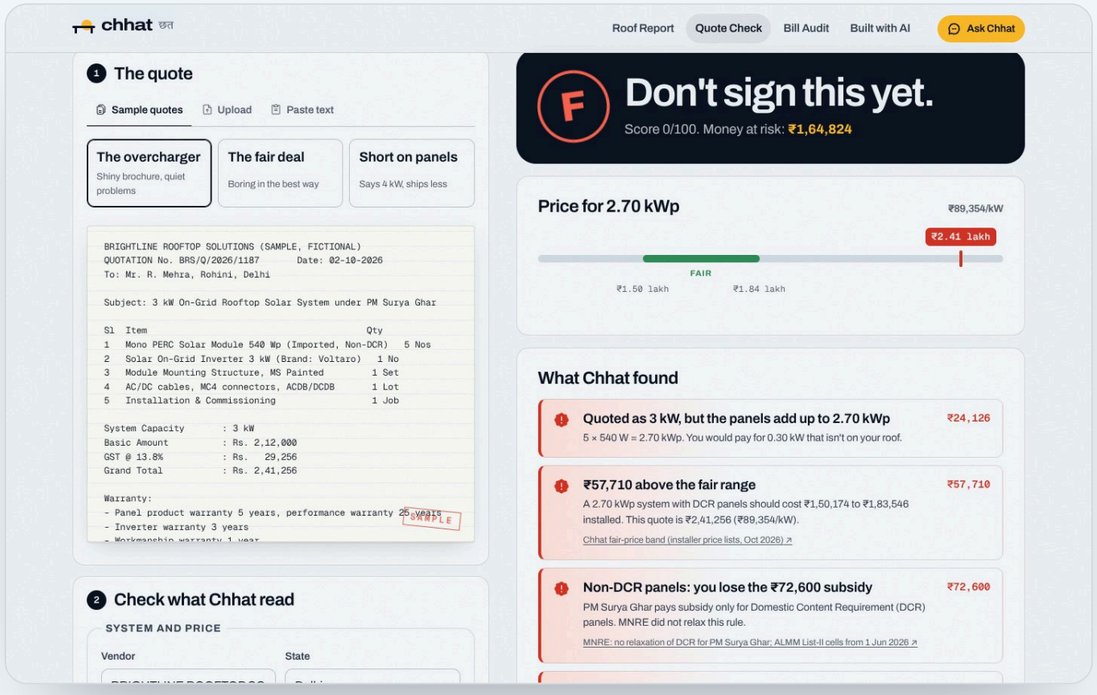
Ask Chhat

Agent in Hindi, Hinglish and English. Every number comes from a tool.

LIVE, NO LOGIN, NO KEY

Works for a stranger in ten seconds.





HOW THE AI WORKS

Claude decides. The engine computes. Nothing is invented.

- **7 typed tools**: irradiance, sizing, subsidy, finance, quote audit, bill audit, sourced knowledge base
- **Two agents, one toolset**: Claude Opus 5.5 on the server; an on-device router when there's no key or no network
- **Document reading**: Claude structured output, or on-device OCR and PDF text, always into an editable form
- **Real data**: NASA POWER (72 cities bundled, live for any lat/lon), MNRE rules, CEA emission factor

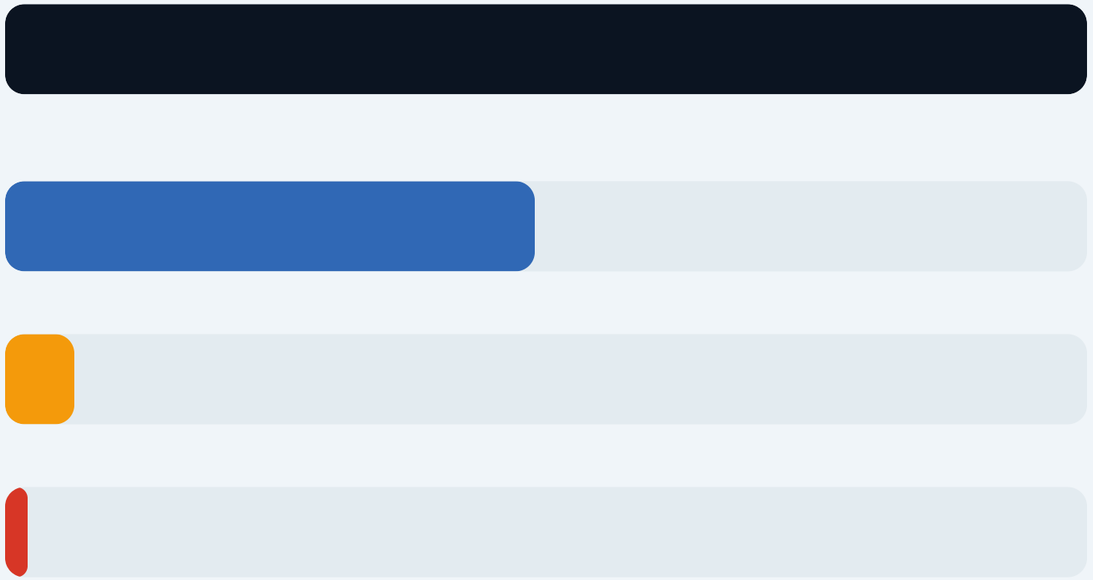
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"Vendor quoted 2.4 lakh for 3 kW in Lucknow, fair?"

→ audit_quote(system_kw: 3, total_price: 240000,
state_code: "UP")

← Grade C · ₹42,000 above fair range · DCR not
stated · no 5-yr warranty

"For 3 kW with DCR panels, a fair installed price is
₹1,62,000 to ₹1,98,000. This quote is about ₹42,000
above that."
```

A ₹1.8 lakh crore installation market, refereed by no one.



₹1,82,400 cr

TAM · installation value of India's 32 GW economic residential rooftop potential (CEEW)

₹89,376 cr

Scheme window · 49 lakh homes left to reach 1 crore

₹11,708 cr

SAM · Chhat's 6.4% take on the TAM

₹87 cr

SOM · FY31 revenue (0.7% of SAM)

BUSINESS MODEL

Free for homes. Paid by the people who want to sell to them.

STREAM	PAYER	PRICE	FY31 (₹ CR)
Verified leads	Installers	₹1,500 / lead	48.0
Success fee	Installers	2% of system value	22.1
Installer SaaS	Installers	₹2,999 / month	9.0
State / DISCOM analytics	Governments	₹40 L / state-year	4.0
Loan origination	Banks, NBFCs	1% of loan	3.6

UNIT ECONOMICS, FY28

₹6,083 contribution on every installed home.

₹224

revenue per report

₹90

variable cost per report (₹85 acquisition + ₹5 AI)

2.5x

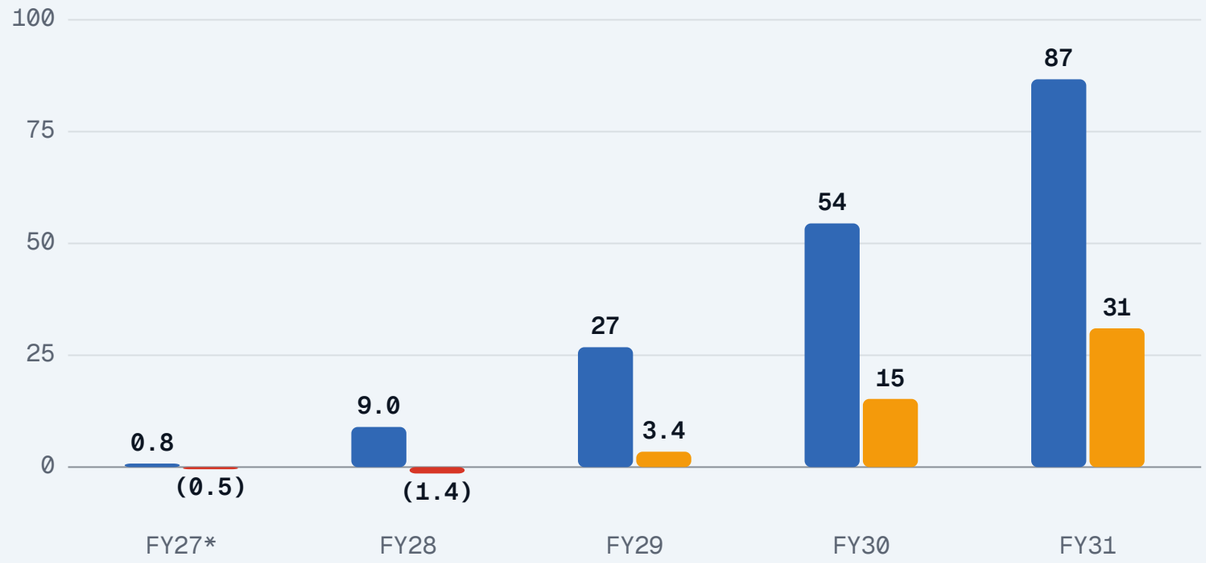
revenue to variable cost

1.6%

of the home's 25-year savings is Chhat's take

Funnel: 8% of reports become verified leads; 20% of leads install. 62.5 reports per installed home.

EBITDA positive in year three.



Revenue

EBITDA

FY31: ₹86.7 cr revenue, 36% EBITDA margin, 150 people. *FY27 is Nov to Mar. Full model with every formula in the repo.

A calculator is easy to copy. Our dataset isn't.

City price index

Every audited quote adds a dated price per kW, per city, per panel type.

DISCOM error map

Every audited bill shows which sub-divisions bill wrong. Regulators will pay for that.

Calibrated yield

Post-install exports tune the PV model city by city, beyond NASA averages.

COMPETITION

Installers sell panels. Chhat referees them.

	INDEPENDENT	AUDITS QUOTES	AUDITS BILLS	HINDI AGENT
SolarSquare, Freyr, Loom	✕	✕	✕	✕
PM Surya Ghar portal	✓	✕	✕	✕
Lead aggregators	✕	✕	✕	✕
Chhat	✓	✓	✓	✓

Start where the subsidy stacks highest.

Nov 2026	Live in Delhi NCR, UP, Maharashtra · first 20 installers
Mar 2027	40,000 reports · first lender · 640 homes influenced
Sep 2027	10 states · “Chhat Verified” badge · first DISCOM pilot
FY28	₹9 crore revenue · 6,400 installs influenced
FY29	EBITDA positive · Series A · C&I and batteries

Growth loop: every report makes a shareable roof card for WhatsApp.

AI-FIRST EXECUTION

One brief in. A startup out.

12,448
lines of source, all written by Claude Code

34
unit tests on the engine and parsers

25
ideas generated, debated and scored before writing code

0
lines of code written by a human

Research, idea tournament, plan, PV model, 3D world, copy, financial model, this deck: AI. The brief and the review: human. Receipts at /built-with-ai.

₹6 crore seed for 15%.

- **45%** product & engineering
- **30%** growth
- **15%** installer network & lead verification
- **10%** compliance & G&A

Gets us to EBITDA break-even in FY29 with ₹3.8 crore of cash at the low point.

Try it: read your roof at the live link. Every number cites its source.